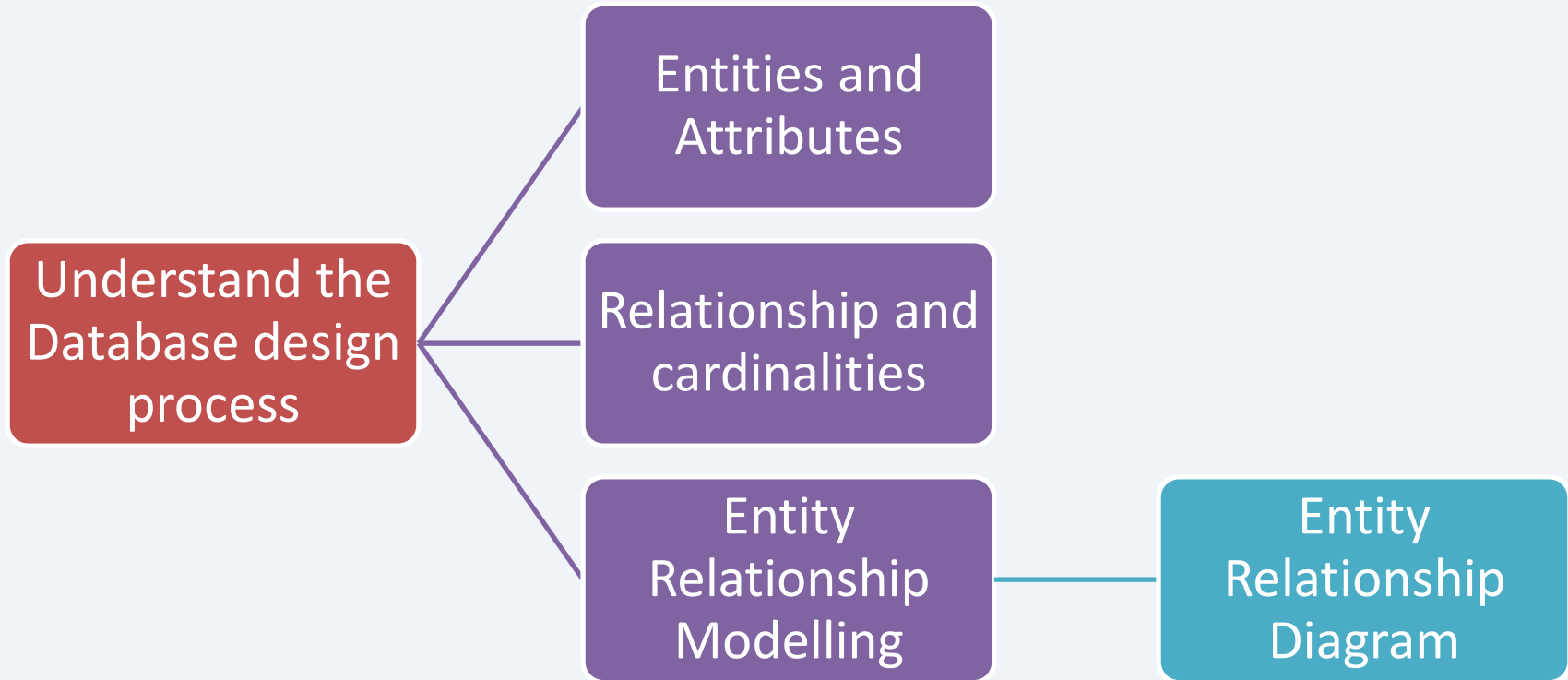


Database Design Principles & Entity Relationship Modelling

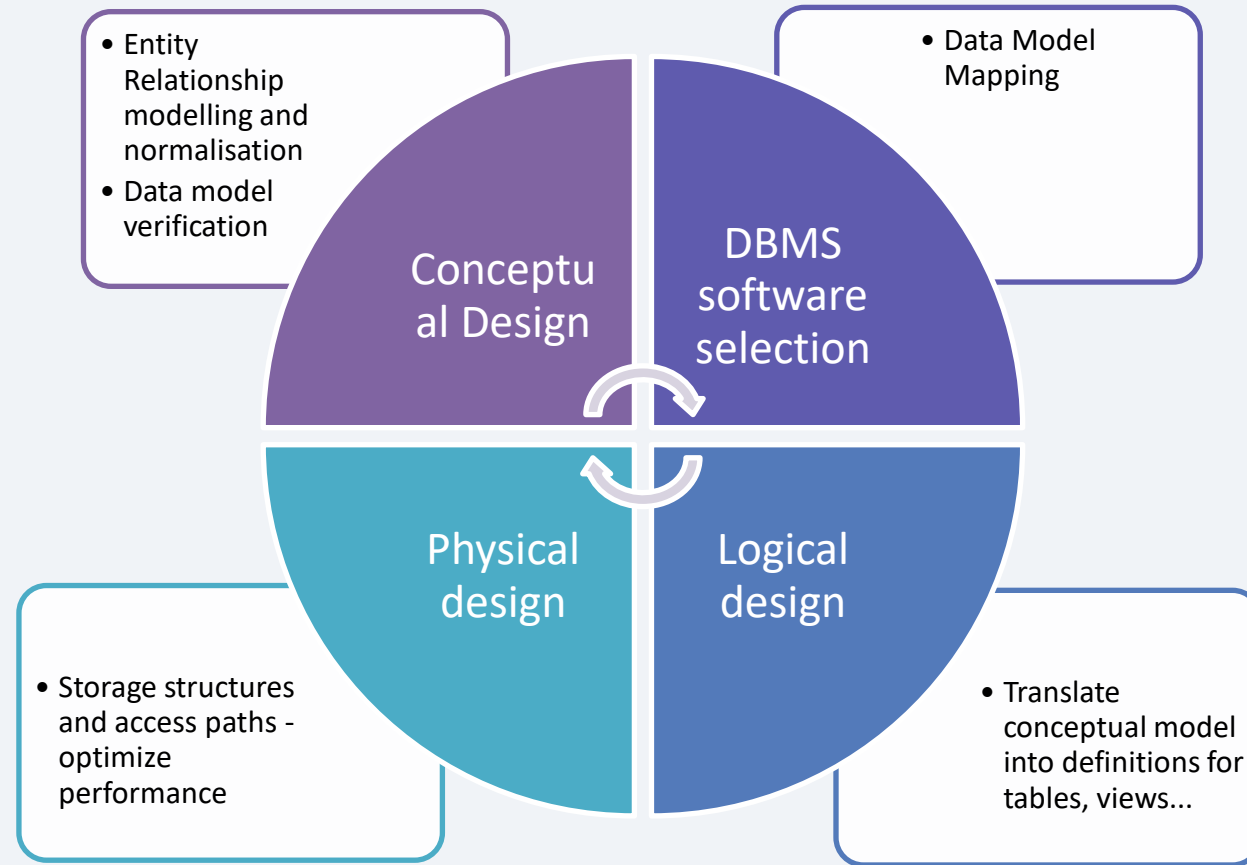
COM 736 Database Systems and Data Analytics

Objectives

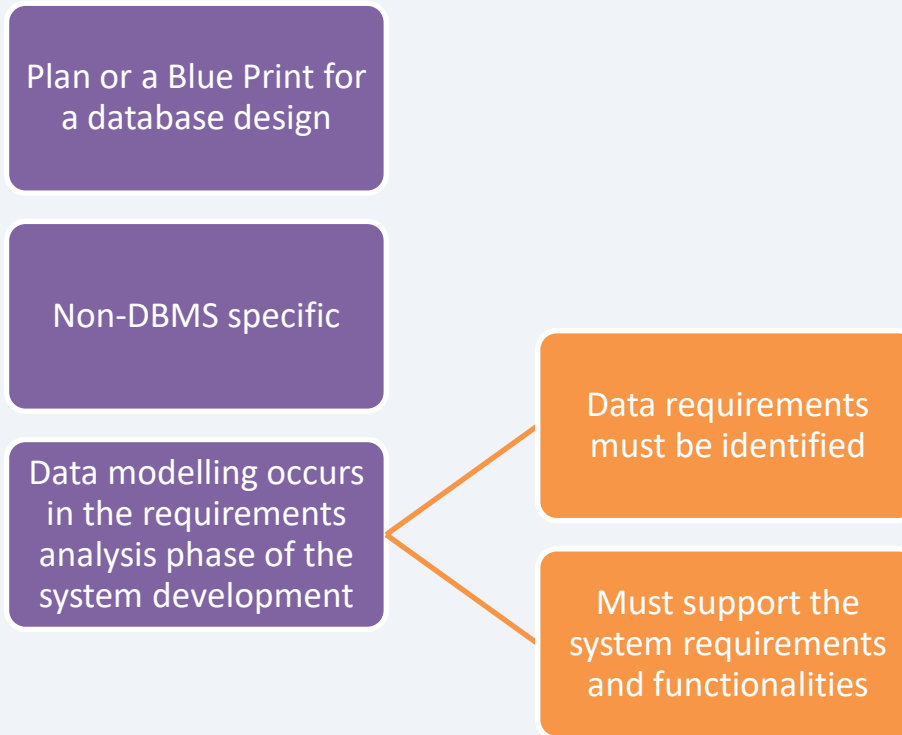


- A Database is an integrated collection of stored operational data used by the application systems of a particular enterprise
- Objectives of the Database
 - To provide the enterprise with **centralised control** of its operational data
 - To provide **data independence**

Database Design



Data Model



Entity Relationship Modelling

Is a diagrammatic technique / design tool

Is a graphical representation of the database system

Attempts to represent the meaning of the real world data

Provides a high- level conceptual data model

Uses entities, attributes and relationships

Requires a proper understanding of the problem

Supports the user's perception of the data

Is DBMS and hardware independent

ER models can then be used to derive tables

Entities

- An entity is any object in the system that we want to model and store information about
- Entities are represented by rectangles (either with round or square corners)

Order

Student

Attribute

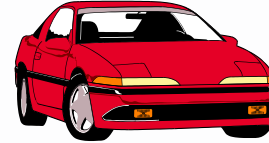
- All the data relating to an entity is held in its attributes.
- An attribute is a property of an entity.
- Each attribute can have any value from its domain.
- Each entity within an entity type:
 - Have the same number of attributes.
 - Every Attribute may not have values.

Entities and attributes

Customer



Car



Attributes

- Name
- Address
- DOB
- Driving licence number

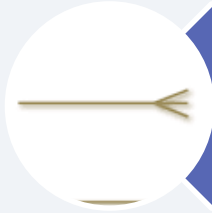
Attributes

- Make
- Model
- Car Reg
- Colour
- Engine Size
- Capacity

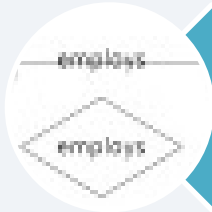
Relationships



A relationship type is a meaningful association between entity types



Relationship types are represented on the ER diagram by a series of lines.



As always, there are many notations in use today...

Types of Entity Relationships: Cardinality

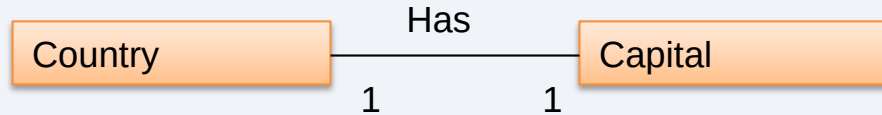
One – to – One

One – to – Many

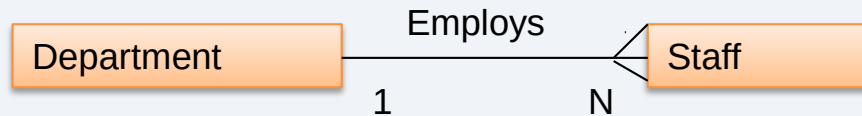
Many – to - Many

Cardinality - Example

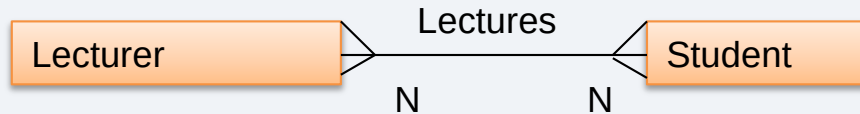
One – To - One



One – To - Many

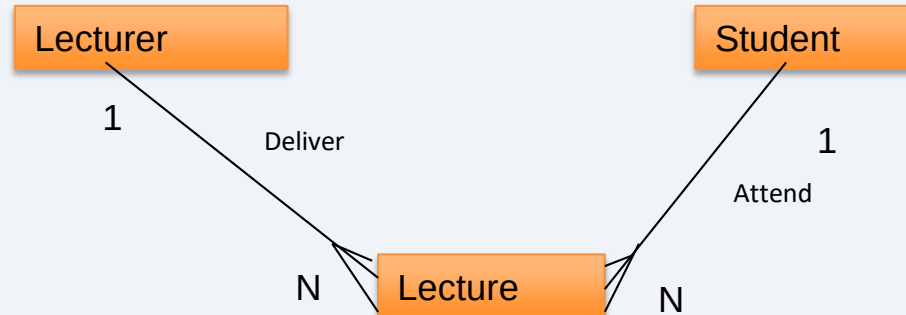
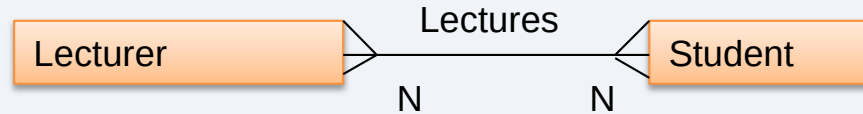


Many – To - Many



Resolve Many-to-Many

Many – To - Many



Next - Example

Practice



Create and Entity Relationship Diagram for a
given University scenario